REDOX 2

17

Transition metals can exhibit more than one oxidation state. To show the oxidation state of the transition metal atom in a compound, roman numerals in brackets are used. For example, copper (II) sulfate means that copper is present in the +2 oxidation state in this compound.

1 Copper (II) sulfate has the formula $CuSO_4$. Work out the oxidation state of sulfur in this compound. Show your working.

2 Give the name of each oxide and work out the oxidation state of chromium.

a CrO b Cr_2O_3 c CrO_3

The oxidation state can help us work out the formula of some compounds of transition metals.

EXAMPLE

i **What is the formula of iron (II) chloride?**

The (II) means that iron is in the +2 oxidation state. Hence the oxidation state of the chlorine atoms present must total –2. Chlorine atoms have the oxidation state of –1 when combined with metals; hence there must be two chlorine atoms present. The formula of iron (II) chloride is $FeCl_2$.

3 Give the formula for each compound.

a iron (III) oxide b cobalt (II) hydroxide

Change in oxidation number Definitions of oxidation and reduction can refer to oxidation number. **Oxidation** is an **increase** in oxidation number. **Reduction** is a **decrease** in oxidation number.

EXAMPLES

ii **Sulfur dioxide and oxygen react to form sulfur trioxide: $2SO_2\ (g) + O_2\ (g) \rightarrow 2SO_3\ (g)$. Has sulfur been oxidised or reduced?**

Study the oxidation states of sulfur in the oxides: In SO_2 sulfur has an oxidation state of +4 (oxygen is 2×-2); in SO_3 sulfur has an oxidation state of +6 (oxygen is 3×-2). The oxidation number of sulfur has changed from +4 to +6. This is an increase, so sulfur has been oxidised.

iii **When zinc metal is added to a solution containing Fe^{3+} ions, solid iron and aqueous Zn^{2+} ions are formed. Which elements have been oxidised and which have been reduced?**

Write the balanced equation with the oxidation numbers underneath:

$$\underset{+3}{2Fe^{3+}\ (aq)} + \underset{0}{3Zn\ (s)} \rightarrow \underset{0}{2Fe\ (s)} + \underset{+2}{3Zn^{2+}\ (aq)}$$

Iron's oxidation number has decreased from +3 to 0; therefore iron has been reduced. Zinc's oxidation number has increased from 0 to +2; therefore zinc has been oxidised. Iron is the oxidising agent (gained electrons) and zinc is the reducing agent (lost electrons).

4 Work out the oxidation numbers of atoms in each reaction then identify which atoms have been oxidised and which have been reduced.

a $2Na\ (s) + CuCl_2\ (s) \rightarrow Cu\ (s) + 2NaCl\ (s)$ b $Zn\ (s) + 2HCl\ (aq) \rightarrow ZnCl_2\ (aq) + H_2\ (g)$

c $SiO_2\ (s) + 2Mg\ (s) \rightarrow Si\ (s) + 2MgO\ (s)$ d $H_2S\ (g) + CuCl_2\ (aq) \rightarrow CuS\ (s) + 2HCl\ (aq)$

e $2KClO_3\ (s) \rightarrow 2KCl\ (s) + 3O_2$ (Hint: in $KClO_3$, chlorine is not bonded to a metal, so the oxidation number is not –1)

We can also use oxidation numbers to help balance reactions.

EXAMPLE

iv **When iron filings are added to a solution of copper (II) sulfate, copper metal is deposited and iron (III) sulfate solution is formed. Write a balanced equation for this reaction.**

Copper (II) is Cu^{2+}. The sulfate ion has the formula SO_4^{2-}, hence copper (II) sulfate has the formula $CuSO_4$. Iron (III) is Fe^{3+}. Hence iron (III) sulfate has the formula $Fe_2(SO_4)_3$, $(2 \times +3 = 3 \times -2)$.
The equation is: $2Fe\ (s) + 3CuSO_4\ (aq) \rightarrow 3Cu\ (s) + Fe_2(SO_4)_2\ (aq)$.

5 Write a balanced equation for each of these redox reactions.

a When copper wire is placed in a solution of silver nitrate ($AgNO_3$), silver metal and copper (II) nitrate are formed.

b Titanium (IV) chloride forms titanium metal and sodium chloride when heated with sodium.

c Lead metal forms lead (II) chloride and hydrogen when reacted with concentrated hydrochloric acid.

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Reduction and metal ores Most metals occur combined with non-metals as compounds, such as carbonates, oxides or sulfides. In order to extract a metal from its ore, the metal has to be reduced. Some of the less reactive metals (e.g. copper, tin and lead) can be obtained from their oxides by heating them with carbon. Copper oxide when heated with carbon gives copper metal and carbon dioxide:
$2CuO\ (s) + C\ (g) \rightarrow 2Cu\ (s) + CO_2$

6 Work out the oxidation numbers for the atoms in the above reaction. Explain why it is clear that the copper has been reduced.

7 When tin (IV) oxide is heated with carbon, tin and carbon dioxide are formed.

a Write a balanced equation for the tin (IV) oxide reaction.

b Show the oxidation states of the atoms in your balanced equation.

Not all metals can be reduced with carbon. Some need a stronger reducing agent such as carbon monoxide (CO). Carbon monoxide can be made by heating a mixture of carbon dioxide and carbon.

8 Write a balanced equation for the synthesis of carbon monoxide.

9 Carbon monoxide reacts with iron (III) oxide to form iron and carbon dioxide.

a Write a balanced equation for the carbon monoxide reaction.

b Show oxidation states of the atoms in your balanced equation.

Although carbon and carbon monoxide are the most common industrial reducing agents, other substances can be used in the laboratory. These include aluminium, hydrogen and methane.

Powdered aluminium can also be used to reduce iron (III) oxide in the **thermit** reaction. Thermit (also spelled thermite) describes a powdered mixture used in welding. Considerable heat is produced by this reaction.

10 Write a balanced equation for the thermit reaction and show the oxidation states of the atoms.

Figure 1 shows how hydrogen can be used to reduce a metal oxide. The general equation is:

metal oxide + hydrogen → metal + water

Hydrogen will only reduce metals found below it in the **Electrochemical** (Reactivity) series.

11 Write a balanced equation for the reduction of copper (II) oxide by hydrogen.

12 Methane (CH_4) can reduce lead (IV) oxide to lead, carbon dioxide and water. Write a balanced equation for this reaction.

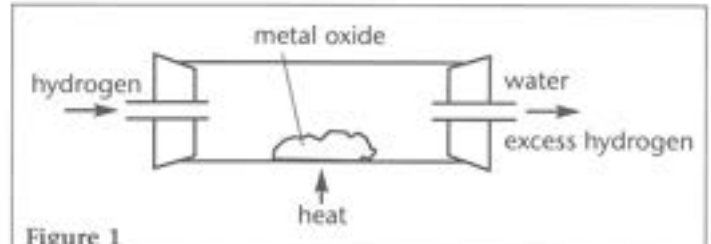


Figure 1

Electrolysis Some metals can only be commercially obtained from their ores by the use of electrolysis. This is the chemical breakdown of compounds by an electric current. For electrolysis, the compound must be in liquid form, either dissolved in water or molten. An electrolytic cell (see figure 2) has two electrodes placed in the compound. When the electric current is switched on, mobile ions are attracted to the electrode of opposite electrical charge. (Remember, 'opposites attract'.)

Aluminium oxide (Al_2O_3) is a solid with a high melting point and has Al^{3+} and O^{2-} ions present. When it is heated and forms a liquid these ions become mobile. In an electrolytic cell, Al^{3+} ions are attracted to the negative electrode and O^{2-} ions are attracted to the positive electrode. At the negative electrode, Al^{3+} ions gain electrons and form Al atoms. At the positive electrode, O^{2-} ions give up electrons and form oxygen atoms.

13 Write a balanced equation for the reaction at the negative electrode where aluminium ions gain electrons and are reduced.

14 Write a balanced equation for the reaction at the positive electrode where oxygen ions lose electrons and are oxidised.

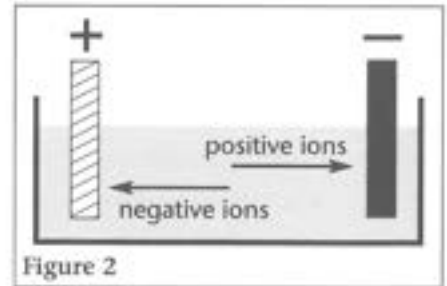


Figure 2

Your answers to questions 13 and 14 are called **half equations** (or half reactions). They can be added (and balanced) to give the complete reaction.

15 Write a balanced equation for the electrolysis of aluminium oxide forming aluminium and oxygen.

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